Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Select *Strengthen* for the statement that would, if true, most strengthen the argument, and select *Weaken* for the statement that would, if true, most weaken the argument. Make only two selections, one in each column.

| Strengthen | Weaken | |
|---|---|---|
| | | None of the genetic differences that have been identified in the genomes of coastal-plain swamp sparrows and freshwater swamp sparrows affect plumage color. |
| | | Mud in tidal marshes tends to be grayish because of the presence of iron sulfide, whereas freshwater mud is browner because of the presence of iron oxide. |
| | | Some species of birds that live in tidal marshes do not have gray plumage. |
| | | The diets of both coastal-plain and freshwater swamp sparrows can change significantly from season to season. |
| | | Baby birds of the coastal-plain subspecies and baby birds of a freshwater swamp subspecies, all raised on an identical diet under controlled conditions, grew plumage similar in color to that of their respective parents. |

•

Boppo is a game played by 2 teams for a fixed duration (number of minutes). It is played with 8 members of each team on the floor—actually playing the game—at all times. At any time during the game, exactly one player from each team must be off the floor—not actually playing. Thus, by taking a break, by entering the game after it starts, or by leaving before it finishes, every player on each team must spend exactly 5 minutes resting or otherwise not playing.

In the table, identify a number of players per team and a number of minutes per game that are consistent with the given information about Boppo. Make only two selections, one in each column.

| Number of players | Number of minutes | |
|---|---|---|
| | | 9 |
| | | 11 |
| | | 18 |
| | | 30 |
| | | 45 |
| | | 50 |

•

Loan X has a principal of $\$x$ and a yearly simple interest rate of 4%. Loan Y has a principal of $\$y$ and a yearly simple interest rate of 8%. Loans X and Y will be consolidated to form Loan Z with a principal of $\$(x+y)$ and a yearly simple interest rate of $r\%$, where $r = \dfrac{4x+8y}{x+y}$. Select a value for $x$ and a value for $y$ corresponding to a yearly simple interest rate of 5% for the consolidated loan. Make only two selections, one in each column.

| $x$ | $y$ | Value |
|---|---|---|
| | | 21,000 |
| | | 32,000 |
| | | 51,000 |
| | | 64,000 |
| | | 81,000 |
| | | 96,000 |

•

Perry, Maria, and Lorna are painting rooms in a college dormitory. Working alone, Perry can paint a standard room in 3 hours, Maria can paint a standard room in 2 hours, and Lorna can paint a standard room in 2 hours 30 minutes.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Perry, Maria, and Lorna have decided that, to speed up the work, 2 of them will paint a standard room together.

Select the value closest to the shortest time in which a 2-person team could paint a standard room, and select the value closest to the longest time in which a 2-person team could paint a standard room, with each person working at his or her respective rate. Make only two selections, one in each column.

| Shortest time | Longest time | |
|---|---|---|
| | | 49 minutes |
| | | 1 hour 7 minutes |
| | | 1 hour 12 minutes |
| | | 1 hour 14 minutes |
| | | 1 hour 22 minutes |
| | | 1 hour 45 minutes |

•

A dance show will include exactly five pieces, and each piece will be presented only once: Requiem, Smooth Step, Trampoline, Unleashed, and Waltzy. Requiem will be presented immediately before Smooth Step. Unleashed will be presented earlier than Trampoline, with at least one piece in between. Waltzy will not be the final piece presented.

In the table, select a piece that cannot be the second piece presented and a piece that cannot be the third piece presented. Make only two selections, one in each column.

| Cannot be presented second | Cannot be presented third | |
|---|---|---|
| | | Requiem |
| | | Smooth Step |
| | | Trampoline |
| | | Unleashed |
| | | Waltzy |

•

An American museum of East Asian art reviewed its acquisitions strategy in 2009. As part of that review, a marketing executive recommended purchasing more ceramics, jades, and paintings of people or landscapes because visitor surveys indicated that those types are of greatest interest to museumgoers in the region. The museum's curator recommended purchasing more twenty-first-century East Asian art—particularly abstract (nonrepresentational) paintings and sculptures—because the museum's collection is weakest in that area.

Among the East Asian works of art described below, select a painting and a sculpture, each of which satisfies at least one of the recommendations made by the marketing executive and at least one of the recommendations made by the curator. Make only two selections, one in each column.

| Painting | Sculpture | |
|---|---|---|
| | | *Painting of girl's face, in shades of blue and red*, acrylic and oil on canvas, 2006 |
| | | *Chinese brush painting of fish in a tank*, ink on paper, 2002 |
| | | *Fish-shaped vase with dragon head*, jade, 1849 |
| | | *Abstract sequence of interconnected curves*, ceramic, 2008 |
| | | *Fossil-like, jagged sculpture suspended from cables*, stainless steel, aluminum, fiberglass, and cable, 2004 |

•

Professor A: The aid industry should begin to focus its efforts to spending on primary schools in the poorest areas, to providing medicines and other basic supplies for health such as mosquito nets, and to a few key agricultural initiatives.

Professor B: Much education work has been ineffective. A village or town with poor schooling may be better off getting a road than a teacher. Once local farmers can transport produce to market, they will be willing to pay for

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

schools—and to make sure the schools succeed.

Suppose that the professors' statements express their genuine opinions. Select statements (1) and (2) as follows: Professor A would likely disagree with (1), and Professor B would take (2) to present logical support for (1). Select only two statements, one per column.

| (1) | (2) | |
|---|---|---|
| | | The aid industry should focus less on the areas of health and agriculture than it now does. |
| | | The aid industry should focus more on primary education than it now does. |
| | | The aid industry should focus its spending less on primary education than it now does. |
| | | Projects in health and agriculture are more likely to be successful if they are not paid for by the aid industry. |
| | | Projects in education are more likely to be successful if they are paid for by the aid industry. |
| | | Projects in education are more likely to be successful if they are paid for by local people. |

•

A car is traveling on a straight stretch of roadway, and the speed of the car is increasing at a constant rate with respect to time. At time 0 seconds, the speed of the car is $v_0$ meters per second; 10 seconds later, the front bumper of the car has traveled 125 meters, and the speed of the car is $v_{10}$ meters per second.

Select values of $v_0$ and $v_{10}$ that are together consistent with the information provided. Make only two selections, one in each column.

| $v_0$ | $v_{10}$ | |
|---|---|---|
| | | 5 |
| | | 18 |
| | | 20 |
| | | 36 |
| | | 72 |

•

For each positive integer $n$, the quantity $s_n$ is defined such that $s_{n+2} = (s_n)^2 - s_{n+1}$. In addition, $s_2 = 1$.

Select values for $s_1$ and $s_4$ that are jointly compatible with these conditions. Select only two values, one in each column.

| $s_1$ | $s_4$ | |
|---|---|---|
| | | −12 |
| | | −7 |
| | | −3 |
| | | −1 |
| | | 0 |

•

The *finesse* of a gold bar is the weight of the gold present in the bar divided by the total weight of the bar. A jeweler has three gold bars. The three bars have identical weights, and the first two bars have identical finesse. If all three bars were melted and combined into one bar, the finesse of the resulting bar would be 0.96.

Select a finesse of the *First two bars* and a finesse of the *Third bar* that are jointly consistent with the given information. Make only two selections, one in each column.

| First two bars | Third bar | |
|---|---|---|
| | | 0.92 |
| | | 0.93 |
| | | 0.94 |

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

| | | |
|---|---|---|
| | | 0.94 |
| | | 0.98 |
| | | 0.99 |

•

A fair coin has 2 distinct flat sides—one of which bears the image of a face and the other of which does not—and when the coin is tossed, the probability that the coin will land face up is $\frac{1}{2}$. For certain values of $M$, $N$, $p$, and $q$, when $M$ fair coins are tossed simultaneously, the probability is $p$ that all $M$ coins land face up, and when $N$ fair coins are tossed simultaneously, the probability is $q$ that all $N$ coins land face up. Furthermore, $M < N$ and $\frac{1}{p} + \frac{1}{q} = 72$.

In the table, select a value for $M$ and a value for $N$ that are jointly consistent with the given information. Make only two selections, one in each column.

| $M$ | $N$ | |
|---|---|---|
| | | 2 |
| | | 3 |
| | | 4 |
| | | 5 |
| | | 6 |
| | | 7 |

•

Educational policy analyst: To improve the long-term economic benefits that our country's universities provide to their students, the government should provide subsidies enabling the universities to hire more academic staff. Hiring more academic staff would allow smaller class sizes. Statistically, university students in smaller classes tend to receive higher grades than those in larger classes. And on average, a student who earns higher grades in university classes tends to have a higher salary after graduation than a student who earns lower grades. Thus, hiring more academic staff nationwide should improve the economic well-being of graduates nationwide.

Consider the incomplete statement:

According to the analyst's argument, students'___1___ is bolstered by their ___2___.

Select for *1* and *2* the options that complete the statement in the manner that most accurately reflects the given information. Make only two selections, one in each column.

| 1 | 2 | |
|---|---|---|
| | | tendency to earn higher grades in university classes |
| | | economic well-being after graduating from university |
| | | being in larger than average university classes |
| | | receiving better education overall |
| | | learning from a greater number of academic staff |

•

The *payload rating* (PR) of a truck is the truck's recommended load weight, which is specified as a number of tonnes (t), where 1 tonne =1,000 kilograms. A certain truck's PR is a whole number of tonnes. The truck has hauled exactly 7 loads, exactly 3 of which had a greater weight than the truck's PR. The weights of these 7 loads, in tonnes, are as follows:

50, 51, 52, 52, 54, 54, 56.

Select the *Least* and the *Greatest* possible values for the truck's PR, in tonnes. Make only two selections, one in each column.

| Least | Greatest | |
|---|---|---|

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

| Least | Greatest | |
|---|---|---|
| | | 49 |
| | | 50 |
| | | 51 |
| | | 52 |
| | | 53 |
| | | 54 |

•

Each of the species of a certain type of insect has at least one of five significant traits: A, B, C, D, and E. Furthermore, one study has determined that any species with Trait A has at least one of the Traits B and C. Another study has determined that any species with Trait C has at least one of the Traits D and E. And a third study has determined that any of the species with Trait B has Trait C. The results of each study are correct.

Select different options for $X$ and for $Y$ such that the following statement most accurately describes the passage. Make only two selections, one in each column.

In addition to the relationship between Traits B and C that is stated explicitly with respect to the third study, the passage implies that any of the insects that has Trait ____$X$____ also has Trait ____$Y$____.

| $X$ | $Y$ | |
|---|---|---|
| | | A |
| | | B |
| | | C |
| | | D |
| | | E |

•

Events A, B, and C have the following probabilities of future occurrence:

A: 20%

B: 50%

C: 80%

On the basis of these probabilities, select for *Both A and B* a description that must be true of the probability that both A and B will occur. And select for *B or C or Both* a description that must be true of the probability that B or C or both will occur. Make only two selections, one in each column.

| Both A and B | B or C or Both | |
|---|---|---|
| | | Must be equal to 40% |
| | | Must be equal to 50% |
| | | Must be equal to 70% |
| | | Must be less than or equal to 20% |
| | | Must be greater than or equal to 80% |

•

Premiolex Corporation spokesperson: In our survey of Premiolex customers, the majority of respondents rated our services as “Excellent,” whereas in our competitor Cretazole's survey, the majority of their respondents rated Cretazole's services as “Very good.” That's proof that—on average, at least—our customers have a higher opinion of Premiolex's services than Cretazole's customers have of its services.

Skeptic: I don’t think the spokesperson's conclusion necessarily follows from the evidence he cites. What if, for example, the Premiolex survey asked participants to choose from among options that weren’t all identical to the ones used in Cretazole's survey?

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Select for *Premiolex survey* and *Cretazole survey* the options such that, in combination, the Premiolex survey and Cretazole survey would support the skeptic's position and be consistent with the spokesperson's report of the results but indicate that the spokesperson's conclusion does not necessarily follow. Make only two selections, one in each column.

| **Premiolex Survey** | **Cretazole Survey** | **Condition: 1,000 respondents were asked to choose among these survey options** | **Results: Number of respondents who chose each survey option** |
|---|---|---|---|
| | | Excellent | 517 |
| | | Average | 482 |
| | | Below average | 1 |
| | | Poor | 0 |
| | | Unacceptable | 0 |
| | | Excellent | 463 |
| | | Average | 501 |
| | | Below average | 0 |
| | | Poor | 35 |
| | | Unacceptable | 1 |
| | | Excellent | 454 |
| | | Average | 511 |
| | | Below average | 34 |
| | | Poor | 0 |
| | | Unacceptable | 1 |
| | | Excellent | 512 |
| | | Average | 478 |
| | | Below average | 10 |
| | | Poor | 0 |
| | | Unacceptable | 0 |
| | | Excellent | 23 |
| | | Average | 516 |
| | | Below average | 230 |
| | | Poor | 231 |
| | | Unacceptable | 0 |

DO NOT COPY filippo.fiz@uptoten.it

•

A satellite is currently in a circular orbit with radius 32,714 km about the center of the earth. (Note: The circumference of a circle is given by $C = 2\pi r$, where $r$ is the radius of the circle and, to the nearest 0.01, $\pi \approx 3.14$ .)

Select for *0.5 km* the increase, to the nearest whole kilometer, in the distance the satellite travels about the center of the earth during each revolution if the orbital radius of the satellite is increased by 0.5 km. And select for *1.5 km* the increase, to the nearest whole kilometer, in the distance the satellite travels about the center of the earth during each revolution if the orbital radius is increased by 1.5 km. Make only two selections, one in each column.

| **0.5 km** | **1.5 km** | |
|---|---|---|
| | | 3 |
| | | 9 |
| | | 27 |
| | | 81 |

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

| | | |
|---|---|---|
| | | 150 |
| | | 450 |

**Questions 451 to 466 — Difficulty: Hard**

451. Companies A and B are part of the same industry and are located in the same city. For Company A, the average (arithmetic mean) salary of its employees, in United Arab Emirates dirhams (AED), is 10,000 AED higher than that for Company B. However, more than half of the employees at Company A have salaries below the average for Company B.

Statement: If the average salary at Company B is ___1___, then the median salary at Company A is ___2___.

Select for *1* and for *2* the options that complete the statement so that it most accurately reflects the information provided. Make only two selections, one in each column.

| 1 | 2 | |
|---|---|---|
| | | (A) greater than 100,000 AED |
| | | (B) less than 100,000 AED |
| | | (C) equal to 110,000 AED |
| | | (D) between 100,000 and 110,000 AED |
| | | (E) greater than 110,000 AED |

452. The following statements describe certain characteristics of a certain pool of candidates for a position. Any candidate who did not meet the minimum qualifications for the position was immediately excluded from consideration. The two candidates who met the minimum qualifications for the position and met all of the desired qualifications also had multiple recommendations. All candidates who received a telephone interview also had extensive experience. All candidates who had extensive experience and impressed the hiring committee during the telephone interview were invited to interview on-site. At least one candidate declined an invitation for an on-site interview, and exactly one candidate was interviewed on-site without receiving a telephone interview.

Consider the following incomplete sentence:

If any candidate ___1___, then that candidate ___2___.

Select for *1* and for *2* two different options that best complete the sentence such that it can be logically inferred from the information provided. Make only two selections, one in each column.

| 1 | 2 | |
|---|---|---|
| | | did not meet the minimum qualifications |
| | | had multiple recommendations |
| | | had extensive experience |
| | | impressed the hiring committee during the telephone interview |
| | | interviewed on-site |

453. In Country C, some but not all eligible voters are required to vote. The particulars of the country's laws governing voting are as follows.

Every citizen who is eligible must vote on Election Day.

A person is eligible if (and only if) that person meets the age requirement and is either a citizen or meets the residency requirement for noncitizens.

The age requirement is that every voter must be at least 19 years old on Election Day.

The residency requirement for noncitizens is that the voter must have been a resident of Country C for at least 5 years on Election Day.

Consider the following individuals:

Abigail: a citizen who is currently 19 years old

Barbara: a 7-year resident noncitizen who is currently 19 years old

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Charles: a 7-year resident noncitizen who is currently 18 years old

For an election held today, select the individual or individuals who must vote based on the information provided, and select the individual or individuals who must not vote based on the information provided. Make only two selections, one in each column.

| Must vote | Must not vote | |
|---|---|---|
| | | Abigail only |
| | | Barbara only |
| | | Charles only |
| | | Abigail and Barbara only |
| | | Abigail and Charles only |
| | | Barbara and Charles only |

454. Most of the cubicles in the north wing of the Acme Company headquarters are less than 20 meters from the nearest fire escape, whereas half of the cubicles in the east wing are not. In other words, those same east-wing cubicles and ___1___ half the north-wing cubicles are ___2___ 20 meters from the nearest fire escape.

Select for *1* and for *2* two options to create a sentence that most accurately restates the information given in the first sentence. Make only two selections, one in each column.

| 1 | 2 | |
|---|---|---|
| | | at least |
| | | at most |
| | | exactly |
| | | greater than |
| | | less than |

455. In June 1990, a small academic press published an initial run of 2,500 copies of linguist Chloe Vermeulen's first book, *Speech and Speaking*. Because the first run was selling well, a second, larger run was produced in June 1995. Total sales for the period from June 1995–June 2000 represented an increase of 52 percent over total sales for the preceding five years; by June 2000, Vermeulen's book had sold a total of 3,843 copies. In 2000, *Speech and Speaking* was cited in an influential paper; subsequently, for the period June 2000–June 2005, sales of Vermeulen's book were double the sales for the previous five-year period.

In the table, identify the number of books that most closely approximates the total sales of *Speech and Speaking* for each of the five-year periods June 1990–June 1995 and June 2000–June 2005, based on the information given. Make only two selections, one in each column.

| June 1990–June 1995 | June 2000–June 2005 | Total sales for the period (number of books) |
|---|---|---|
| | | 1,320 |
| | | 1,530 |
| | | 2,530 |
| | | 2,640 |
| | | 3,800 |
| | | 4,640 |

456. Miguel's online banking password is nine characters long and includes at least one character of each of five types: digits, punctuation marks, uppercase (capital) letters, lowercase (noncapital) letters, and certain other characters. The password begins or ends with one of these other characters, of which it contains exactly one. No consecutive characters in the password are of the same one of these types. (For example, no capital letter is preceded or followed by another capital letter.) The password contains more lowercase letters than characters of any one of the other four types. It also contains fewer punctuation marks than digits, lowercase letters, or uppercase letters. The first three characters of the password are M, ?, and G.

In the table, select a type of character that the fifth character in the password must be and a type of character that

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

In the table, select a type of character that the fifth character in the password must be and a type of character that the seventh character must be. Make only two selections, one in each column.

| Fifth character | Seventh character | |
|---|---|---|
| | | Digit |
| | | Lowercase letter |
| | | Punctuation mark |
| | | Uppercase letter |
| | | Other character |

457. Public health professional: The Happydale Department of Environmental Services (DES) has inspected area beaches for pathogenic bacteria for over 20 years. Currently, DES monitors 50 freshwater beaches and 10 coastal beaches on a weekly basis. Program participation by owners of beaches that are open to the public is voluntary. Beach owners who choose not to participate in the program are allowing the public to recreate on entirely unmonitored beaches. The DES now proposes adoption of the Adopt-a-Beach Program, designed to promote health and environmental education as well as public involvement in the protecting of public beaches. The program would require DES to collect biological data from all area beaches. It will also help reduce the number of beach advisories, whereby the public must be warned of dangerous conditions on certain beaches.

On the basis of the public health professional's statement, indicate in the table which of the following is most likely to result if the DES proposal is successfully adopted, and indicate which is most likely to result if the DES proposal is not adopted. Make only two selections, one in each column.

| Adopted | Not adopted | |
|---|---|---|
| | | Incidents of waterborne disease will increase. |
| | | The number of beach advisories will increase. |
| | | DES will shift its principal priorities from monitoring to education. |
| | | There will be fewer, if any, entirely unmonitored beaches. |
| | | Some beaches will remain entirely unmonitored. |

458. Marco: Giant kangaroos—one of several extinct species of large mammals (megafauna)—went extinct in Australia around 46,000 years ago. The deposits and wear patterns on the teeth of these animals from around the time of their extinction indicate that they fed mostly on saltbrush shrubs. Saltbrush thrives in arid climates, so it is not likely that the kangaroos' food supply was adversely affected by the increasing aridity of the climate at the time. Thus, something else would have to account for their extinction, and the best candidate for that cause is predation by humans.

Fatima: That argument alone is not likely to satisfy many researchers in this field. Have you found any other evidence to bolster your conclusion?

Select *Marco* for the statement that, if true, most justifies Marco's assertions, and select *Fatima* for the statement that, if true, most justifies Fatima's skepticism about Marco's assertions. Make only two selections, one in each column.

| Marco | Fatima | |
|---|---|---|
| | | Giant kangaroos became extinct during a period that was less arid than previous periods they endured. |
| | | Many researchers believe humans first arrived in Australia around 40,000 years ago. |
| | | Approximately 60 different species in Australia died out in the wave of extinctions around 46,000 years ago. |
| | | Fossils of giant kangaroos also show evidence that those animals' diets routinely included plants other than saltbrush. |
| | | Several types of megafauna larger than the giant kangaroo went extinct around 46,000 years ago. |

459. A mattress company has two stores, one in City X and the other in City Z. The company has advertised equally in newspapers in both cities but has advertised twice as much on the radio in City Z as in City X. The two cities

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

in newspapers in both cities but has advertised twice as much on the radio in City Z as in City X. The two cities have similar populations and economies, and the sales at each store have been roughly equal. A consultant claims this shows that the radio advertising has not improved mattress sales. In the table below, select changes that the company could make in City X and City Z, respectively, that together would probably be most helpful in testing the consultant's claim. Make only two selections, one in each column.

| City X | City Z | |
|---|---|---|
| | | Double newspaper advertising |
| | | Eliminate newspaper advertising |
| | | Eliminate radio advertising |
| | | Change the content of radio advertising |
| | | Add television advertising |

460. Metro Ballet Company presents high-quality productions of traditional, classical ballet. For the past several years, however, the company's overall profits have been declining, and ticket sales have been flat. Annual audience surveys indicate that a majority of those who attend Metro Ballet productions consistently enjoy the performances and prefer classical ballet to other forms of dance; almost all of them have been attending Metro Ballet for several years. General surveys of area residents indicate, however, that very few are aware of Metro Ballet productions, and most imagine that the performances are boring and the tickets too expensive. In an effort to appeal to a wider audience, over the past decade the company has spent increasing amounts of money on spectacular stage productions, while lowering ticket prices.

In the first column, select the strategy that, in the absence of the other alternatives listed, would lead most directly to decreasing Metro Ballet's expenses for its classical ballet productions. In the second column, select the strategy that, in the absence of the other alternatives listed, would constitute the most direct approach to solving the problem of increasing audience size for Metro Ballet's classical ballet productions. Make only two selections, one in each column.

| Decrease expenses | Increase audience size | |
|---|---|---|
| | | Obtain public funding to double the spending on stage productions without increasing ticket prices |
| | | Return spending on productions to levels of several years ago |
| | | Expand productions to include modern, folk, and tap dance traditions |
| | | Offer special discounts to reward people who have attended the greatest number of performances |
| | | Mount a local advertising campaign emphasizing the affordability and excitement of Metro Ballet's spectacular stage productions |

461. At XYZ Inc. any employee receives a verbal warning upon accumulating at least 3 unexcused absences within any 365-day period and a written reprimand upon accumulating at least 4 such absences. For any single 8-hour workday, missing between 10 minutes and 2 hours of work counts as one-third of an absence, missing between 2 hours and 4 hours of work counts as half an absence, and missing more than 4 hours counts as a full absence. However, an employee may stay late to make up for up to 1 hour of an unexcused absence on the same day.

The following options are descriptions of the unexcused absences of 5 employees of XYZ Inc. Assume that in each case the employee had no other unexcused absences and made up no other time. Select a description of an employee who qualified for a verbal warning but not a written reprimand, and select a description of an employee who qualified for a written reprimand. Make only two selections, one in each column.

| Verbal warning | Written reprimand | |
|---|---|---|
| | | Absent all day on 5 April 2010, 8 June 2010, 17 April 2011, and 14 June 2011 |
| | | Absent 4.5 hours but stayed 1 hour late on 13 May 2010; absent all day on 2 June 2010, 1 May 2011, and 21 July 2011 |
| | | Absent 4.5 hours on 19 March 2010; stayed one hour late on 20 March 2010; absent all day on 8 February 2011 and 9 February 2011; arrived 40 minutes late on |

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

| | | |
|---|---|---|
| | | absent all day on 8 February 2011 and 9 February 2011; arrived 40 minutes late on 17 April 2011 |
| | | Absent 3.5 hours on 13 September 2010; absent 1 hour on 15 September 2010; absent 6 hours on 16 September 2010; absent 2.5 hours on 18 September 2010; absent 1 hour on 19 September 2010 |
| | | Absent 3 hours on 7 July 2010; absent 2.5 hours on 13 September 2010; absent all day on 31 January 2011 and 4 July 2011; absent 5 hours on 12 March 2011 |

462. Marketing strategist: Agency A designed an advertising campaign that our company is about to test with a focus group. We are wondering whether a new ad campaign will increase our name recognition among consumers. As a contingency, we have decided that we might ask Agency B to design an alternate campaign. However, if we find that Agency A's campaign elicits positive responses from the focus group, we will not ask Agency B for a campaign.

A statement that must be true if the marketing strategist's statements are true: After the focus testing is complete, if the company ___1___, then it must also be the case that the company ___2___.

Select for 1 *if* and 2 *then* the two different options that create a statement that must be true if the marketing strategist's statements are true. Make only two selections, one in each column.

| 1 if | 2 then | |
|---|---|---|
| | | believes Agency A's campaign will increase the company's name recognition |
| | | believes Agency A's campaign will decrease the company's name recognition |
| | | asks Agency B for a campaign |
| | | asks Agency A for an alternate campaign |
| | | believes the focus group did not respond positively to Agency A's campaign |

463. Francois and Pierre each owe Claudine money. Today, Francois will make a payment equal to 50% of the amount he owes Claudine, and Pierre will make a payment equal to 10% of the amount he owes Claudine. Together, the two payments will be equal to 40% of the combined amount that Francois and Pierre owe Claudine.

Select for *Francois* and *Pierre* amounts that Francois and Pierre could owe Claudine that are jointly consistent with the given information. Make only two selections, one in each column.

DO NOT COPY filippo.fiz@uptoten.it